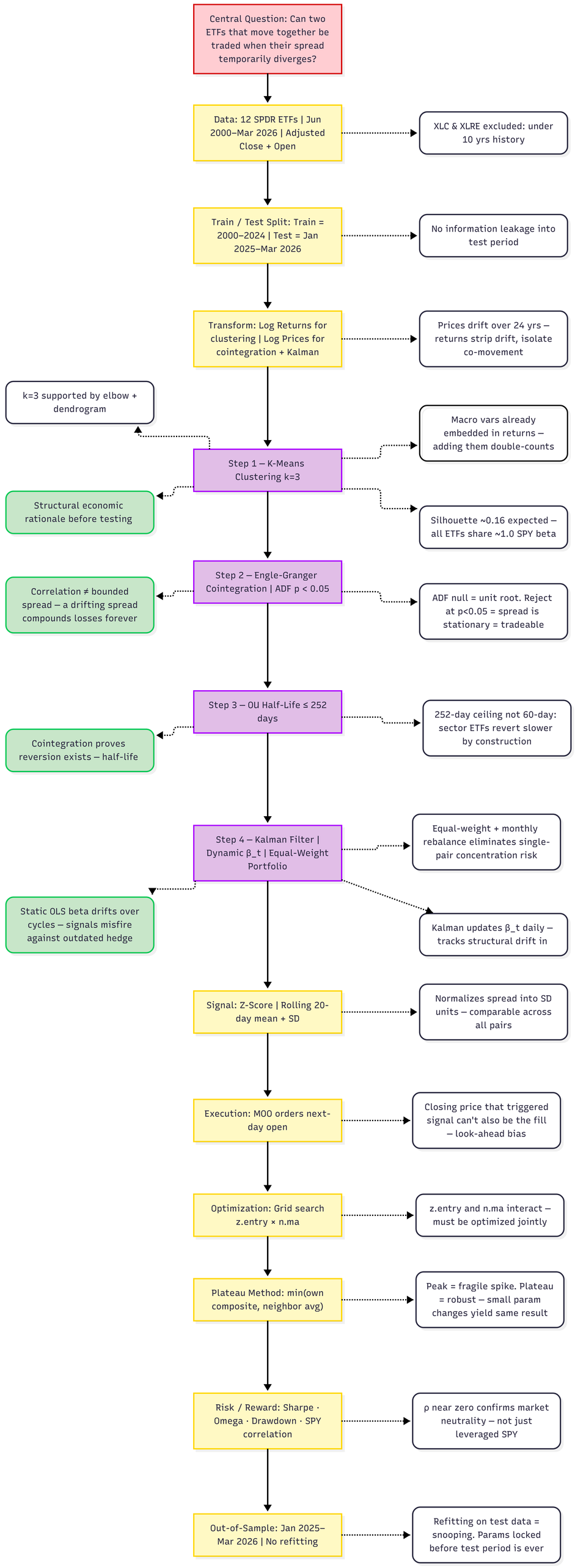
Central Question: Can two ETFs that move together be traded when their spread temporarily diverges?
Data: 12 SPDR ETFs | Jun 2000–Mar 2026 | Adjusted Close + Open
XLC & XLRE excluded: under 10 yrs history
Train / Test Split: Train = 2000–2024 | Test = Jan 2025–Mar 2026
No information leakage into test period
Transform: Log Returns for clustering | Log Prices for cointegration + Kalman
Prices drift over 24 yrs – returns strip drift, isolate co-movement
k=3 supported by elbow + dendrogram
Macro vars already embedded in returns – adding them double-counts
Step 1 – K-Means Clustering k=3
Structural economic rationale before testing
Silhouette ~0.16 expected – all ETFs share ~1.0 SPY beta
Step 2 – Engle-Granger Cointegration | ADF p < 0.05
Correlation ≠ bounded spread – a drifting spread compounds losses forever
ADF null = unit root. Reject at p<0.05 = spread is stationary = tradeable
Step 3 – OU Half-Life ≤ 252 days
252-day ceiling not 60-day: sector ETFs revert slower by construction
Cointegration proves reversion exists – half-life
Step 4 – Kalman Filter | Dynamic β_t | Equal-Weight Portfolio
Equal-weight + monthly rebalance eliminates single-pair concentration risk
Static OLS beta drifts over cycles – signals misfire against outdated hedge
Kalman updates β_t daily – tracks structural drift in
Signal: Z-Score | Rolling 20-day mean + SD
Normalizes spread into SD units – comparable across all pairs
Execution: MOO orders next-day open
Closing price that triggered signal can't also be the fill – look-ahead bias
Optimization: Grid search z.entry × n.ma
z.entry and n.ma interact – must be optimized jointly
Plateau Method: min(own composite, neighbor avg)
Peak = fragile spike. Plateau = robust – small param changes yield same result
Risk / Reward: Sharpe · Omega · Drawdown · SPY correlation
ρ near zero confirms market neutrality – not just leveraged SPY
Out-of-Sample: Jan 2025–Mar 2026 | No refitting
Refitting on test data = snooping. Params locked before test period is ever